Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

$n = 3$

$c = [8, 3, 1, 2]$

There are 3 ways to make change for $n = 3$: $\{1, 1, 1\}$, $\{1, 2\}$, and $\{3\}$.

Function Description

Complete the getWays function in the editor below.

getWays has the following parameter(s):

- int n: the amount to make change for
- int c[m]: the available coin denominations

Returns

- int: the number of ways to make change

Input Format

The first line contains two space-separated integers n and m, where:

n is the amount to change

m is the number of coin types

The second line contains m space-separated integers that describe the values of each coin type.

Constraints

- $1 \leq c[i] \leq 50$
- $1 \leq n \leq 250$
- $1 \leq m \leq 50$
- Each $c[i]$ is guaranteed to be distinct.

Hints

Solve overlapping subproblems using Dynamic Programming (DP):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the overlapping subproblems. Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for 0 cents? - How many ways can you make change for > 0 cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case $(n = 0)$.

- The answer may be larger than a 32-bit integer.

Sample Input 0

```
4 3
1 2 3
```

```
6  import re
7  import sys
8
9  #
10 # Complete the 'getWays' function below.
11 #
12 # The function is expected to return a LONG_IN
13 # The function accepts following parameters:
14 #  1. INTEGER n
15 #  2. LONG_INTEGER_ARRAY c
16 #
17
18 def getWays(n, c):
19     dp = [0] * (n + 1)
20
21     dp[0] = 1
22
23     for coin in c:
24         for amount in range(coin, n + 1):
25             dp[amount] = dp[amount] + dp[amoun
26
27     return dp[n]
28 if __name__ == '__main__':
29     fptr = open(os.environ['OUTPUT_PATH'], 'w'
30
31     first_multiple_input = input().rstrip().sp
32
```

Line: 46 Col: 1

Upload Code as File | Run Code | Submit Code

Test against custom input

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin) Download

```
4 3
1 2 3
```

Your Output (stdout)

```
4
```

Expected Output Download

```
4
```